Patient Medical Record

Longitudinal chart, February 2024 - September 2026

Notice: every name, number, date, and clinical detail in this document is invented. It exists so you can practice chunking, embedding, and retrieval (RAG) on realistic clinical text. Do not use it for medical decisions.

1. Patient Demographics

Field	Value
Full name	Karim BENALI
Medical record number (MRN)	00482917
Date of birth / Age	12 April 1971 (55 years at last visit)
Sex	Male
Occupation	Secondary-school mathematics teacher
Marital status	Married, 3 children
Smoking status	Former smoker, 15 pack-years, quit in 2019
Alcohol	None
Primary care physician	Dr. Nadia Cherif, Internal Medicine
Cardiologist	Dr. Yacine Boudiaf
Emergency contact	Salima Benali (spouse)

2. Active Problem List

Problem	Onset / Diagnosed	Status
Type 2 diabetes mellitus	2019	Active, controlled (HbA1c 6.9% in March 2026)
Essential hypertension	2021	Active, controlled on losartan
Hyperlipidemia	2021	Active, LDL at goal on high-intensity statin
Mild non-proliferative diabetic retinopathy	January 2025	Active, annual eye exam
Coronary artery disease, single-vessel (mid-LAD)	September 2025	Treated with drug-eluting stent; on dual antiplatelet therapy
Obesity (BMI 31.4 in Feb 2024, 28.9 in Sept 2026)	2018	Improving with diet and exercise

3. Allergies and Intolerances

Agent	Reaction	Type	Recorded
Penicillin	Diffuse skin rash with itching	Allergy	2009
Lisinopril (ACE inhibitor)	Persistent dry cough	Intolerance (not a true allergy)	May 2024

Family history: father died of myocardial infarction at age 58; mother has type 2 diabetes; one brother with hypertension.

4. Clinical Encounter Notes

14 February 2024 - Annual primary care visit

Provider: Dr. Nadia Cherif

Chief complaint: Routine yearly check-up; patient reports occasional morning headaches and increased thirst over the last two months.

Vitals: BP 148/92 mmHg, heart rate 78 bpm, weight 92 kg, BMI 31.4, temperature 36.7 C.

Assessment: Hypertension above target. Diabetes poorly controlled with HbA1c 8.1%. LDL 142 mg/dL, above target for a patient with diabetes.

Plan: Start lisinopril 10 mg once daily. Continue metformin 1000 mg twice daily. Start atorvastatin 20 mg nightly. Dietitian referral. Recheck labs and blood pressure in 3 months.

20 May 2024 - Follow-up visit

Provider: Dr. Nadia Cherif

Interval history: Patient reports a persistent dry cough beginning about three weeks after starting lisinopril. No fever, no shortness of breath, chest X-ray not required.

Vitals: BP 136/84 mmHg, heart rate 74 bpm, weight 89 kg.

Assessment: ACE-inhibitor cough. HbA1c improved to 7.4%. LDL improved to 96 mg/dL.

Plan: Stop lisinopril and record as an intolerance. Start losartan 50 mg once daily. Continue metformin and atorvastatin 20 mg. Recheck potassium and creatinine in 2 weeks.

09 January 2025 - Ophthalmology consultation

Provider: Dr. Amel Kaci, Ophthalmology

Findings: Dilated fundus exam shows scattered microaneurysms in both eyes. No macular edema. Visual acuity 10/10 in both eyes.

Impression: Mild non-proliferative diabetic retinopathy, both eyes.

Plan: No laser treatment needed. Repeat dilated exam in 12 months. Emphasize glucose and blood pressure control.

03 August 2025 - Emergency department visit

Provider: Dr. Rachid Mansouri, Emergency Medicine

Chief complaint: Chest pressure for 20 minutes while climbing stairs, relieved by rest. Two similar episodes in the previous 2 weeks.

Vitals: BP 152/90 mmHg, heart rate 88 bpm, oxygen saturation 98% on room air.

Workup: 12-lead ECG: normal sinus rhythm, no ST changes. High-sensitivity troponin negative at 0 and 3 hours. Chest X-ray normal.

Assessment: Possible stable angina; acute coronary syndrome ruled out at this time.

Disposition: Discharged home with sublingual nitroglycerin 0.4 mg as needed and aspirin 81 mg daily. Outpatient exercise stress test arranged.

21 August 2025 - Exercise stress test with imaging

Provider: Dr. Yacine Boudiaf, Cardiology

Result: Patient exercised 7 minutes (Bruce protocol) and stopped for chest tightness. ST depression of 2 mm in leads V4-V6. Perfusion imaging showed a reversible defect in the anterior wall (LAD territory). Ejection fraction 58%.

Impression: Positive for inducible ischemia.

Plan: Coronary angiography scheduled.

5. Hospital Admission: Coronary Angiography and PCI

Admission 10 September 2025 - Discharge 11 September 2025

Procedure: Coronary angiography via right radial access. Findings: 80% stenosis of the mid left anterior descending (LAD) artery. Right coronary and circumflex arteries had no significant disease (less than 30% narrowing). One drug-eluting stent (3.0 x 18 mm) was placed in the mid LAD with excellent result and no residual stenosis.

Complications: None. Radial access site clean, no hematoma.

Discharge medications: aspirin 81 mg daily, clopidogrel 75 mg daily for 12 months (do not stop without calling cardiology), atorvastatin increased to 80 mg nightly, metoprolol succinate 50 mg daily, losartan 50 mg daily, metformin 1000 mg twice daily, nitroglycerin 0.4 mg sublingual as needed.

Instructions: avoid heavy lifting for 1 week, begin cardiac rehabilitation in 4 weeks, follow-up with cardiology in 6 weeks.

12 March 2026 - Primary care follow-up

Provider: Dr. Nadia Cherif

Interval history: No chest pain since the stent. Completed cardiac rehabilitation, walks 40 minutes daily. Reports mild fatigue with metoprolol in the first month, now resolved.

Vitals: BP 124/76 mmHg, heart rate 62 bpm, weight 84 kg, BMI 29.6.

Assessment: HbA1c 6.9%. LDL 62 mg/dL, at target of below 70. Kidney function normal (eGFR 82).

Plan: Add empagliflozin 10 mg daily for cardiovascular and glycemic benefit. Counsel on genital hygiene and hydration. Continue all other medications.

15 September 2026 - Cardiology and primary care review

Provider: Dr. Yacine Boudiaf and Dr. Nadia Cherif

Interval history: Stent placed 12 months ago. No angina, no palpitations. One episode of mild genital yeast infection in May 2026, treated with topical clotrimazole.

Vitals: BP 122/74 mmHg, heart rate 60 bpm, weight 82 kg, BMI 28.9.

Assessment: Stable CAD, excellent control. HbA1c 6.6%. LDL 58 mg/dL. eGFR 78. Retinopathy exam due in January 2027.

Plan: Stop clopidogrel at the 12-month mark (date reached 10 September 2026) and continue aspirin 81 mg indefinitely. Continue atorvastatin 80 mg, metoprolol 50 mg, losartan 50 mg, metformin 1000 mg twice daily, empagliflozin 10 mg. Next review in 6 months.

6. Current Medication List (as of 15 September 2026)

Medication	Dose	Frequency	Indication	Started
Aspirin	81 mg	Once daily	Coronary artery disease, stent	Aug 2025
Atorvastatin	80 mg	Nightly	Hyperlipidemia, CAD	Feb 2024 (20 mg); 80 mg since Sep 2025
Metoprolol succinate	50 mg	Once daily	CAD, heart rate control	Sep 2025
Losartan	50 mg	Once daily	Hypertension	May 2024
Metformin	1000 mg	Twice daily with meals	Type 2 diabetes	2019
Empagliflozin	10 mg	Once daily	Diabetes, cardiovascular protection	Mar 2026
Nitroglycerin SL	0.4 mg	As needed for chest pain	Angina	Aug 2025

Discontinued: Lisinopril 10 mg (Feb-May 2024, stopped for cough). Clopidogrel 75 mg (Sep 2025 - Sep 2026, completed 12-month course).

7. Laboratory Results Over Time

Test (units)	Ref. range	14 Feb 2024	20 May 2024	12 Mar 2026	15 Sep 2026
HbA1c (%)	below 5.7	8.1	7.4	6.9	6.6
Fasting glucose (mg/dL)	70-99	168	132	112	104
LDL cholesterol (mg/dL)	below 70 (high risk)	142	96	62	58
HDL cholesterol (mg/dL)	above 40	38	41	47	49
Triglycerides (mg/dL)	below 150	220	181	139	128
Creatinine (mg/dL)	0.7-1.3	0.9	0.9	0.95	1.0
eGFR (mL/min/1.73 m2)	above 90	92	91	82	78
Potassium (mmol/L)	3.5-5.1	4.3	4.6	4.4	4.5
Urine albumin/creatinine (mg/g)	below 30	34	-	22	18

Note: the mildly elevated urine albumin in February 2024 normalized after blood pressure and glucose control improved. The small decline in eGFR from 92 to 78 is expected after starting empagliflozin and is being monitored.

8. Vital Signs Trend

Date	BP (mmHg)	Heart rate (bpm)	Weight (kg)	BMI
14 Feb 2024	148/92	78	92	31.4
20 May 2024	136/84	74	89	30.4
03 Aug 2025 (ED)	152/90	88	-	-
12 Mar 2026	124/76	62	84	29.6
15 Sep 2026	122/74	60	82	28.9

9. Care Plan and Patient Education Summary

Diet: Mediterranean-style diet, limit refined sugars and salt (under 5 g/day). Dietitian visits completed in 2024 and 2025.

Exercise: 150 minutes of moderate activity weekly. Patient currently walks 40 minutes daily and swims twice a week.

Warning signs requiring urgent care: chest pain lasting more than 10 minutes or not relieved by one nitroglycerin tablet, pain spreading to the arm or jaw, sudden shortness of breath, fainting, or black stools while on aspirin.

Sick-day rule: if vomiting or unable to drink fluids, temporarily hold empagliflozin and metformin and call the clinic.

Upcoming: dilated eye exam in January 2027, cardiology and primary care review in March 2027, annual flu vaccination each autumn.

Appendix: Practice Questions for Your RAG System

Use these to test retrieval. Answers are in the document, so you can check whether your pipeline finds them. Questions are ordered from simple lookup to harder multi-chunk reasoning.

#	Question	Type	Expected answer
1	What is the patient's MRN?	Single lookup	00482917
2	What allergy does the patient have?	Single lookup	Penicillin (skin rash)
3	Why was lisinopril stopped?	Single lookup	Persistent dry cough (ACE-inhibitor intolerance)
4	What medication replaced lisinopril?	Single lookup	Losartan 50 mg daily
5	What did the angiography find?	Single lookup	80% mid-LAD stenosis, treated with one drug-eluting stent
6	How did HbA1c change over time?	Multi-chunk	8.1 to 7.4 to 6.9 to 6.6 percent
7	Which medications was the patient on right after the stent was placed?	Multi-chunk	Aspirin, clopidogrel, atorvastatin 80, metoprolol, losartan, metformin, nitroglycerin
8	Is clopidogrel still active today, and why?	Reasoning	No. The 12-month course ended in September 2026; aspirin continues
9	Why did eGFR decline slightly in 2026?	Reasoning	Expected effect after starting empagliflozin; being monitored
10	What is the patient's cardiovascular family risk?	Multi-chunk	Father died of MI at 58
11	What side effects did empagliflozin cause?	Multi-chunk	One mild genital yeast infection in May 2026
12	What should the patient do if vomiting and unable to drink?	Single lookup	Hold empagliflozin and metformin and call the clinic
13	What is the patient's blood type?	Unanswerable	Not in the document. A good RAG system should say it does not know

Ideas for experimenting

Try different chunk sizes (200, 500, 1000 tokens) and see how question 6 and 7 change, since the answers span several sections. Compare chunking by fixed size against chunking by heading. Add metadata such as encounter date and section name to each chunk, then test date-filtered retrieval like 'what happened in 2025?'. Question 13 checks that your prompt discourages hallucination.